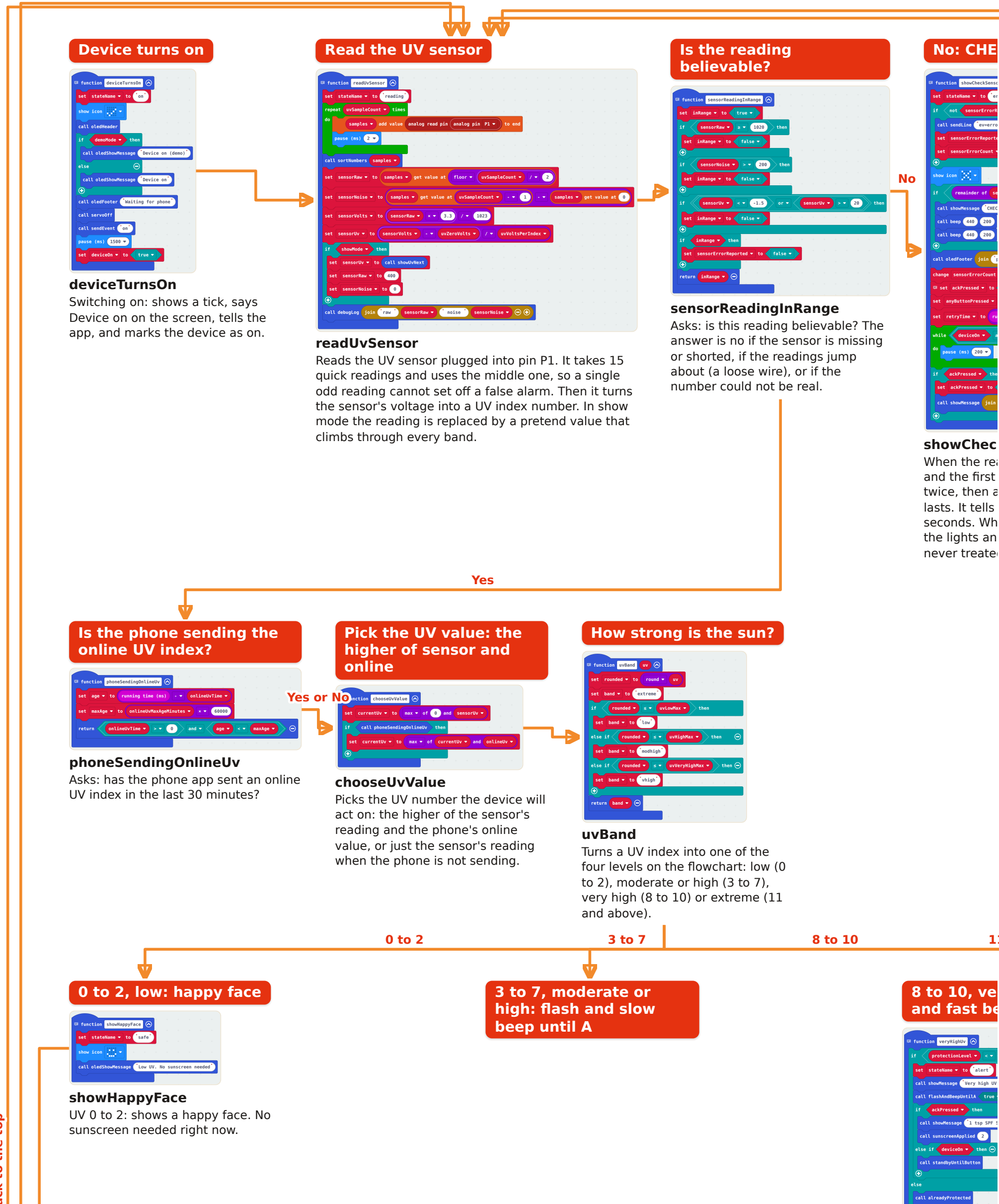


Sunburn device: the program or

The Sunburn Flowchart, built from the real blocks of the program, with what each one does

The Sunburn device is a wearable built on a BBC micro:bit, a small computer with 25 lights and two buttons. It measures the sun's UV where you are through a phone, and then flashes, beeps or taps your arm until you put sunscreen on. Two hours later it reminds you to built from 78 coloured blocks. Every one of them is on these sheets, with a note saying what it does.

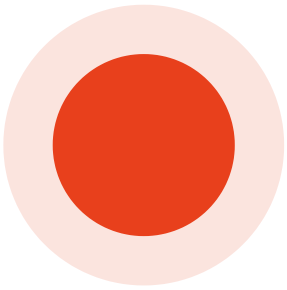
Follow the arrows. Each red label is a box of the flowchart; the coloured picture under it is the actual block of code that does that job, exactly as it looks in the Ma forever block runs this plan round and round, all day.



in the wrist

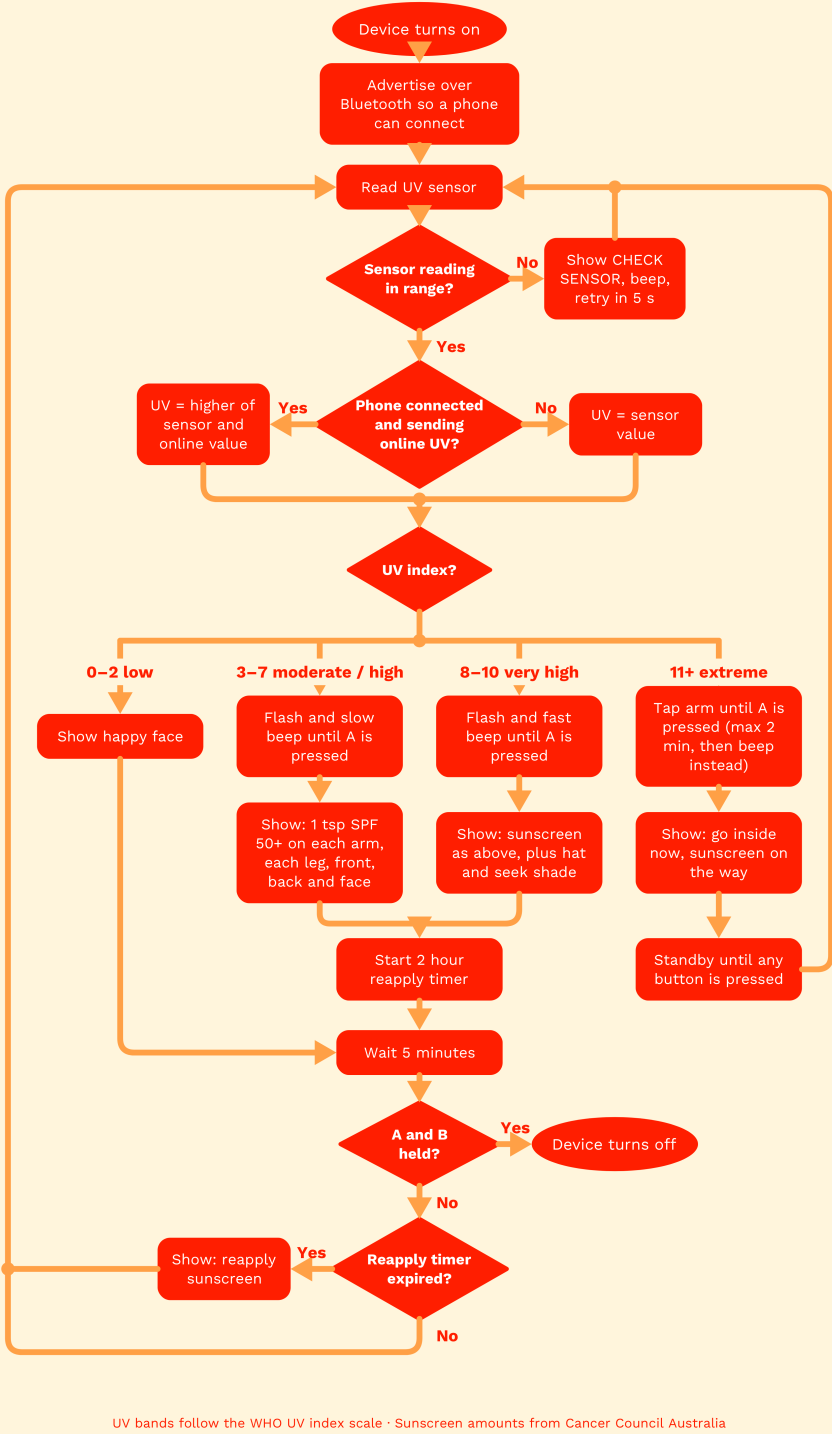
with a sensor, checks the online UV index for
o reapply. The program that does all this is

akeCode editor, and the note says what it does. The



The Sunburn Flowchart

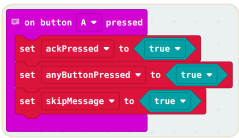
Sunburn Flowchart



The plan the device follows. Every box on it is a block on the left, with the same name.

The two buttons

A: done



on button A pressed
Pressing A means done: sunscreen is on, or I am going inside. It also wakes the device, skips a long message, and during a wait shows the UV number and the countdown.

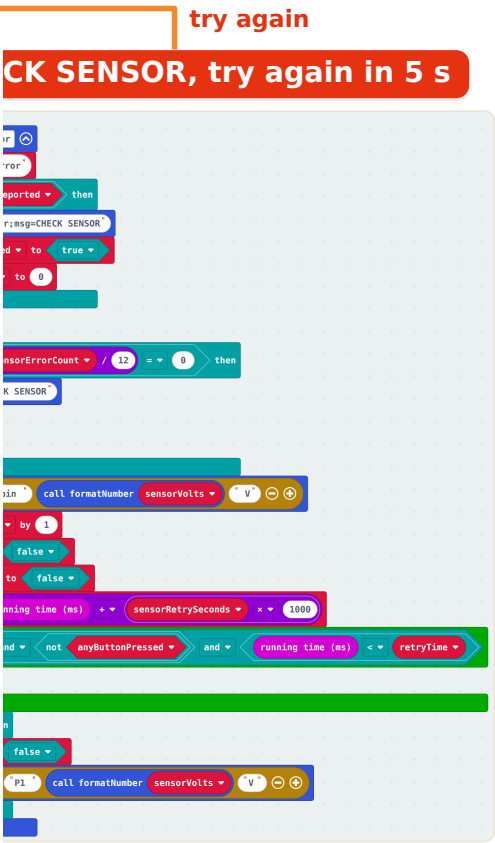
Pressing A and B together turns the device off, or on again (see the flowchart).

The whole program

B: wake up



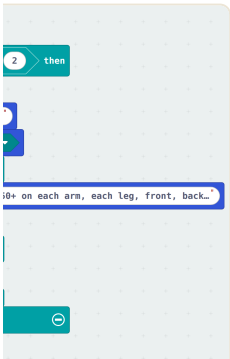
on button B pressed
Pressing B wakes the device from sleep. Holding B while switching on gives the fast demo timings.



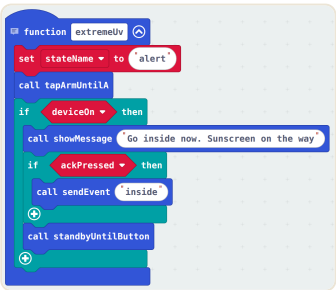
kSensor
ading is not believable: shows a cross, time scrolls CHECK SENSOR and beeps again about once a minute while the fault the app once, and tries again every 5 ile it waits, A shows the pin voltage on d B retries at once. A faulty sensor is d as safe.

11 and above

very high: flash beep until A



11 and above, extreme: tap the arm until A



extremeUv
UV 11 or more: taps the arm until A

**moderateOrHighUv**

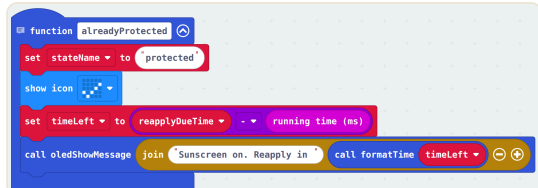
UV 3 to 7: flashes and beeps slowly until A is pressed, then shows how much sunscreen to put on and starts the 2 hour timer. If sunscreen is already on, it just shows a tick and the time left. If nobody presses A within 10 minutes, the device assumes it is not being worn and goes to sleep.

**veryHighUv**

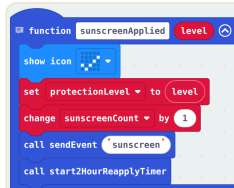
UV 8 to 10: flashes quickly until A is pressed, then shows the sunscreen hat and shade, a 2 hour timer. It alerts the wearer so far on moderate advice.

sunscreen already on A pressed

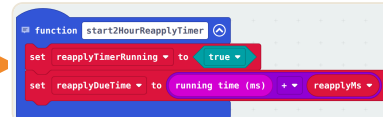
A pressed

Sunscreen already on: just a tick**alreadyProtected**

Sunscreen is already on, so there is no alert: just a tick and the time until it needs reapplying.

A pressed: sunscreen is on**sunscreenApplied**

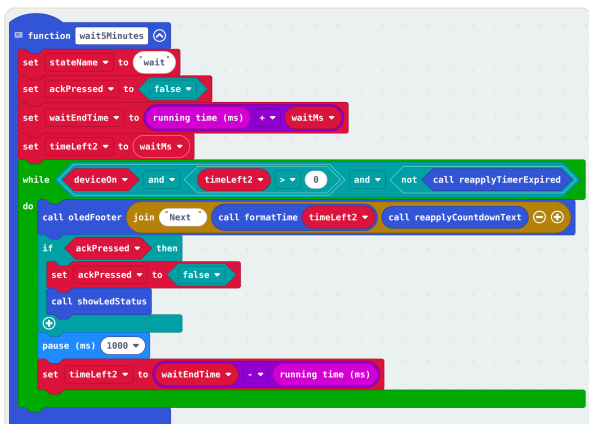
A was pressed after a sunscreen alert: remembers which advice was given, counts one more application, tells the app, and starts the timer.

Start the 2 hour reapply timer**start2HourReapplyTimer**

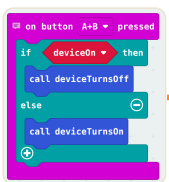
Starts the 2 hour countdown to the next sunscreen.

SL pr**sta**

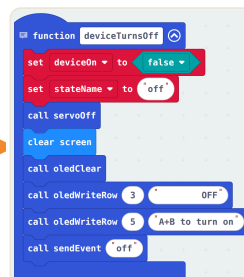
Slee
from
toget
After
read
waki

Wait 5 minutes**wait5Minutes**

Waits 5 minutes before the next check. The wait ends early if the device is turned off or the sunscreen timer runs out. Pressing A during the wait shows the UV number and the countdown.

A and B held?**on button A+B pressed**

Pressing A and B together turns the device off, or on again.

Yes: device turns off**deviceTurnsOff**

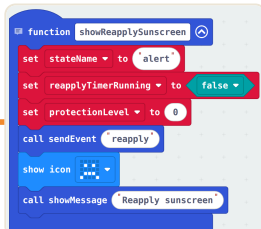
Switching off: everything stops and the screen shows OFF. Bluetooth stays on so the app can switch it back on.

No

Is it time for more sunscreen?**reapplyTimerExpired**

Asks: have the 2 hours since the last sunscreen passed?

Yes

Yes: show Reapply sunscreen**showReapplySunscreen**

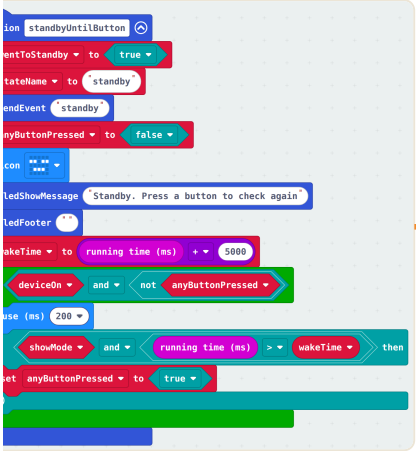
Shows Reapply sunscreen and clears the timer, so the next trip round the flowchart alerts again if the sun is still strong.



ies and beeps
pressed, then
reen advice plus
and starts the 2
erts again if the
ly had the
a.

is pressed (after 2 minutes it beeps
instead), shows Go inside now, then
sleeps until a button is pressed.

leep until any button is
ressed



ndbyUntilButton

p: waits for any button, or done
n the app. Pressing A and B
ether still turns the device off.
rwards it goes straight back to
ling the sensor. In show mode it
es by itself after 5 seconds.



All 78 blocks in the editor, in nine sections. The flowchart on this sheet is
sections 1 to 4; sections 5 to 9 are the helpers: screen, sound, Bluetooth,
maths and the optional screen.

What the block colours mean

- | | |
|-------------------------------|-----------------------------|
| Variables: remember a value | Text |
| Functions: a step with a name | Arrays: lists |
| Basic: the lights, pauses | Music: beeps |
| Input: buttons and time | Pins: sensor, motor, screen |
| Loops: repeat | Bluetooth: the phone |
| Logic: if, compare | Serial: the USB cable |
| Maths | Control: timing |

Words used here

- UV index** How strong the sun's burning rays are, from 0 (none) to 11 and above (extreme). Protection is needed from 3 upwards, even on cool or cloudy days.
- micro:bit** The small computer the program runs on. It has a 5 by 5 grid of lights, buttons A and B, and sockets called pins (P0, P1, P2) that other parts plug into.
- Block** One instruction of the program. Blocks snap together and run in order from top to bottom. A block with a name at the top (function) is a step that other blocks can call on.
- Sensor** The part that measures UV. It gives a voltage, which the program turns into a UV index number.
- Servo** A small motor with an arm. Here it taps the wearer's wrist when the sun is extreme.
- Bluetooth and USB** The two ways the wearable talks to a phone or a laptop. The phone app sends the online UV index; the wearable sends back what it is doing.
- OLED** An optional small screen. Without one, the grid of lights does the job.
- ms and Hz** ms are milliseconds, thousandths of a second. Hz sets how high a beep sounds.
- Demo timings** Much shorter waits (seconds instead of minutes and hours) so the whole day can be shown in a few minutes at a demo.
- Names in the blocks** Words like uvSampleCount are the program's own names for its settings and memory. Read them as English: uv sample count is how many readings to take.

Every one of the 78 blocks, with its note, is in the booklet on the table.